STOCHASTIC OPTIMIZATION OF EV CHARGING STATION ALLOCATION UNDER UNCERTAIN DEMAND

A mini project on deciding where to build chargers when demand is uncertain

Course: Stochastic Decision Making

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Note: The demand values, costs, penalties, and probabilities in this report are assumed for an academic simulation. They are not taken from a real charging network.

Abstract

A city has to decide how many new electric-vehicle (EV) chargers to build in each zone before it knows how many drivers will arrive on a given day. If it builds too few, drivers are turned away or sent to a distant zone. If it builds too many, some chargers stay idle and their cost is wasted. In this project I model this decision for six zones as a two-stage stochastic program. In the first stage, the number of new chargers per zone is chosen under a total budget. In the second stage, after demand is known, each zone's demand is served at its own chargers, sent to a nearby zone at a travel cost, or left unmet at a penalty. Demand uncertainty is represented by 100 Monte Carlo scenarios for optimization, and the chosen plans are then tested on 1,000 new scenarios. Under the assumptions used here, the stochastic plan reduces expected daily cost from 1,661.47 to 1,244.70, a saving of 416.77 (25.1%) compared with a plan built only for average demand. It also raises the service level from 95.2% to 99.0%.

1. Problem and objectives

The main question is: **How many new chargers should be built in each zone so that the expected total cost is lowest when demand is uncertain?**

The project has five practical objectives:

- Define a charger-allocation decision and the cost of serving or missing demand.
- Represent uncertain demand with a set of random scenarios.
- Find the allocation that minimizes install cost plus expected recourse cost.
- Compare it with a deterministic plan built on average demand, and measure the Value of the Stochastic Solution (VSS).
- Check how the result changes with the budget and with the level of demand volatility.

2. Model assumptions

To keep the model manageable, I assume one day of operation for a network of six zones. All values below are illustrative and are not estimated from real data.

- Each charger serves up to 30 charging sessions per day. Existing chargers are fixed and free to use.
- A new charger costs 60 per day (installation cost spread over its life). At most 8 new chargers can be built in one zone and at most 16 in total.
- A session that cannot be served costs a penalty of 25 (lost profit).
- A driver may use a charger in another zone if it is within 6 km. The travel cost is 1.5 per km per session. Distances are straight-line distances.
- Demand in zone j on scenario s is the zone's mean demand multiplied by a city-wide random shock and a local random noise term. Both are lognormal with mean 1, with standard deviations (log scale) of 0.15 and 0.25.

3. Data and mathematical formulation

The six zones, their assumed mean daily demand (sessions per day), map coordinates in km, and existing chargers are:

Zone	Mean demand	X (km)	Y (km)	Existing chargers
Downtown	120	5	5	2
University	90	2	8	1
Mall	100	8	2	1
Airport	70	12	9	1
Residential	80	1	2	1
Tech Park	110	9	8	2
Total	**570**			**8**

The expected total demand is 570 sessions per day, while the existing 8 chargers can serve only 8 × 30 = 240 sessions. New chargers are therefore needed, but demand on any single day can be much higher or lower than the average. Across the 1,000 evaluation scenarios, total daily demand averages 566.9 sessions, and on the busiest 5% of days it exceeds 760.5.

3.1 Decision variables

- $\mathbf{x}_j$: number of new chargers built in zone j before demand is known (integer, 0 to 8). This is the first-stage decision.
- $\mathbf{t}_{ijs}$: sessions from zone i that are served at zone j in scenario s (only allowed if the distance is at most 6 km).

- u_{is}: sessions from zone i that are left unmet in scenario s.

3.2 Second-stage (recourse) problem

Once the demand d is known, the best way to serve it is found by solving a small linear program for the given x:

$$Q(x, d) = \min \sum_{ij} c_{ij} t_{ij} + p \sum_{i} u_i$$
$$\text{subject to } \sum_{j} t_{ij} + u_i = d_i \text{ for every zone } i$$
$$\sum_{i} t_{ij} \le 30 (e_j + x_j) \text{ for every zone } j$$
$$t_{ij} \ge 0, u_i \ge 0$$

Here $c_{ij} = 1.5 \times$ distance between zones i and j (zero when i = j), p = 25 is the unmet-demand penalty, and e_j is the number of existing chargers. For example, if a zone has 3 chargers in total, it can serve 90 sessions. If 120 sessions arrive and no nearby zone has spare capacity, 30 sessions are lost, which costs 30 × 25 = 750.

3.3 Objective

The first-stage decision minimizes the install cost plus the average recourse cost over the S scenarios (the extensive form of the two-stage problem):

$$\min 60 \sum_{j} x_j + (1/S) \sum_{s} Q(x, d_s)$$
$$\text{subject to } \sum_{j} x_j \le 16, x_j \in \{0, 1, \ldots, 8\}$$

The **deterministic baseline** is the same model with only one scenario, the average demand. Comparing the two plans on the same fresh scenarios shows how much value there is in modelling uncertainty.

4. Method followed

I first solved the optimization models and then tested the resulting plans on separate simulated days, so that the comparison is not biased by the scenarios used for optimization.

Step 1 — Generate demand scenarios

A random seed of 42 generates 100 scenarios for optimization. A different seed (43) generates 1,000 new scenarios used only for evaluation.

Step 2 — Build the deterministic plan

The optimization is solved with a single scenario equal to the mean demand of each zone.

Step 3 — Build the stochastic plan

The extensive-form mixed-integer program with the 100 scenarios is solved, with charger counts as integer variables.

Step 4 — Evaluate both plans on fresh scenarios

For each of the 1,000 evaluation scenarios, the second-stage linear program is solved for both plans. The install cost is added to give the total cost of each scenario.

Step 5 — Compute performance measures

The measures are mean cost, standard deviation of cost, 95th percentile of cost (a risk measure), service level (share of sessions served), probability of any unmet demand, and VSS = (expected cost of the deterministic plan) − (expected cost of the stochastic plan).

Step 6 — Check sensitivity

The budget (8 to 24 chargers) and the zone-level volatility (0.1 to 0.4) are varied, and both plans are rebuilt and re-evaluated each time.

Tools used

Python is used for the calculations. NumPy generates the random scenarios, pandas stores results in tables, PuLP with the CBC solver solves the integer program, and SciPy (HiGHS) solves the many second-stage linear programs. The interactive dashboard is built with Streamlit and Plotly, and the figures in this report were drawn with Matplotlib from the same model code. The repository contains the application code, the requirements file, screenshots, and this report.

5. Results

On the 1,000 evaluation scenarios, the stochastic plan has an expected daily cost of **1,244.70**, compared with **1,661.47** for the deterministic plan. The Value of the Stochastic Solution is therefore **416.77** per day, about 25.1% of the deterministic cost. The deterministic plan builds 12 new chargers, which gives a total capacity of 600 sessions against a mean demand of 570. The stochastic plan uses the full budget of 16 new chargers (720 sessions of capacity), adding more chargers in Downtown, University, Airport, and Tech Park.

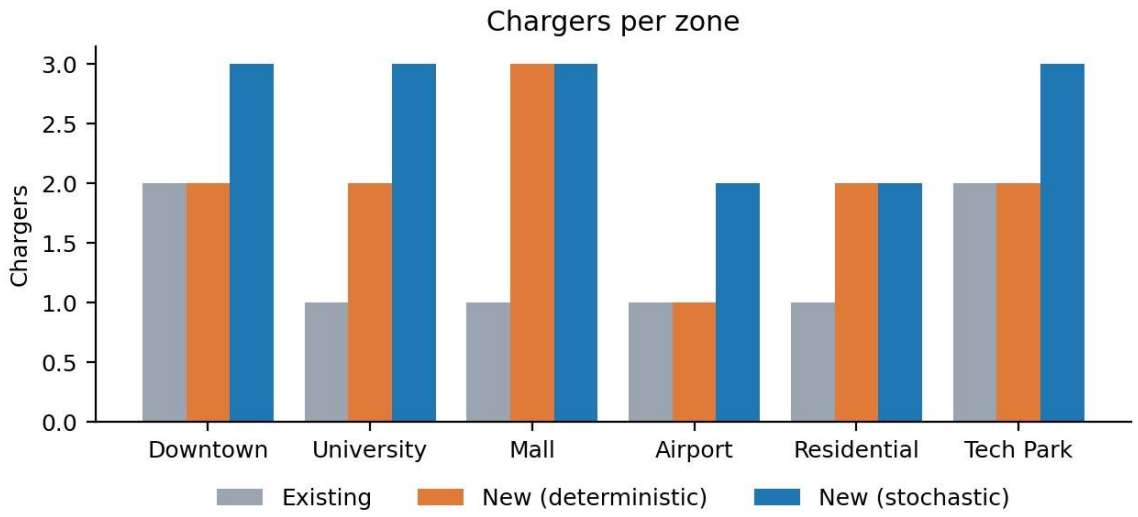


Figure 1. Existing and new chargers per zone for the deterministic and stochastic plans.

Zone	Existing	New (deterministic)	New (stochastic)	Avg unmet / day (det.)	Avg unmet / day (sto.)
Downtown	2	2	3	5.06	1.13
University	1	2	3	4.95	0.73
Mall	1	3	3	3.08	1.09
Airport	1	1	2	5.69	0.48
Residential	1	2	2	3.36	1.33
Tech Park	2	2	3	5.31	0.77
Total	**8**	**12**	**16**	**27.44**	**5.54**

The stochastic plan spends 240 more on installation than the deterministic plan (960 against 720), but the expected cost of transfers and unmet demand falls from 941.47 to 284.70, a saving of about 656.77. The extra chargers therefore pay for themselves under the assumed penalty.

Metric	Deterministic plan	Stochastic plan
Mean daily cost	1,661.47	1,244.70
Install cost	720.00	960.00
Expected transfer + unmet cost	941.47	284.70
Standard deviation of cost	1,427.09	705.23
95th percentile cost	4,897.46	2,496.01
Service level	95.16%	99.02%
Probability of any unmet demand	32.6%	7.9%

The stochastic plan is also less risky. The standard deviation of cost is about half that of the deterministic plan (705.23 against 1,427.09), and the cost on the worst 5% of days is 2,496.01 instead of 4,897.46. Some demand is left unmet on 32.6% of days under the deterministic plan, compared with 7.9% under the stochastic plan.

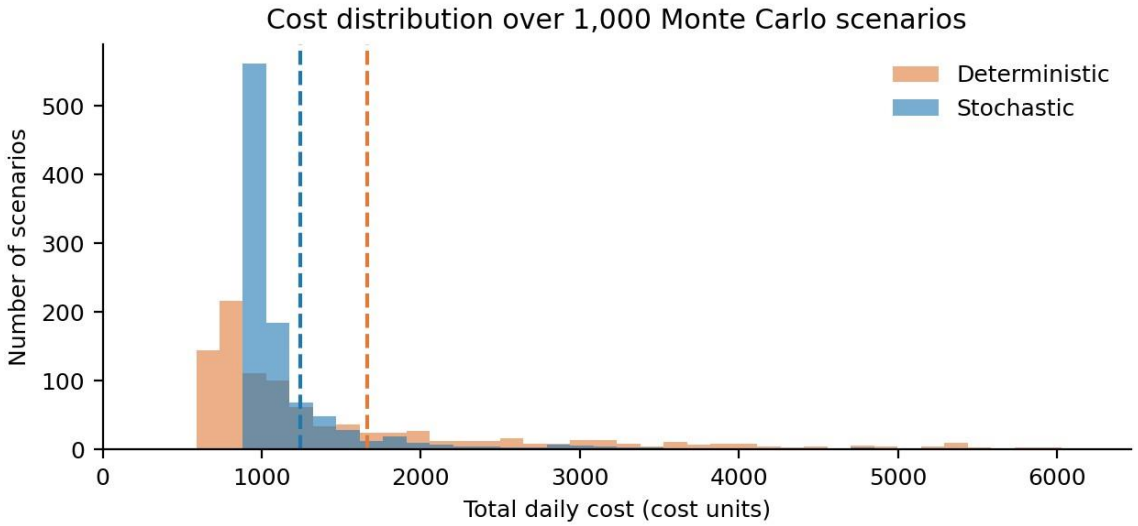


Figure 2. Distribution of total daily cost for both plans (axis cut at the 99th percentile). Dashed lines show the mean cost.

One caution is that the optimization objective on the 100 training scenarios was 1,132.68, which is about 112.02 lower than the 1,244.70 measured on fresh scenarios. This is why the plans were compared on a separate set of scenarios instead of the training set.

6. Simulation and sensitivity analysis

The simulation draws many demand scenarios, solves the second-stage problem for each, and records the cost. Along with the mean, the code records the standard deviation, the 95th percentile, the service level, and the probability of unmet demand. These values describe the spread of outcomes under the model; they are not guarantees about a real network.

6.1 Simulated demand

Zone	Mean	Std. dev.	Minimum	Maximum
Downtown	119.7	36.2	47.0	295.5

University	89.3	26.6	33.1	224.6
Mall	99.7	30.1	34.9	254.4
Airport	69.4	20.5	27.0	161.6
Residential	79.0	23.6	32.0	188.8
Tech Park	109.7	32.4	39.5	275.2

The city-wide shock moves all zones in the same direction on the same day. On a busy day every zone is busy at once, so a neighbouring zone is unlikely to have spare capacity. This is the main reason why planning only for the average is costly.

6.2 Sensitivity to the budget

I rebuilt both plans for budgets of 8, 12, 16, 20, and 24 new chargers. To keep this analysis quick, each stochastic plan used 60 optimization scenarios and was evaluated on 500 new scenarios, so the numbers differ slightly from the main run.

Budget (new chargers)	Deterministic cost	Stochastic cost	VSS	Service level (sto.)	Chargers used (sto.)
8	3,109.29	3,122.94	-13.65	82.7%	8
12	1,688.30	1,688.30	0.00	94.9%	12
16	1,688.30	1,263.70	424.60	99.0%	16
20	1,688.30	1,247.99	440.32	99.4%	17
24	1,688.30	1,247.99	440.32	99.4%	17

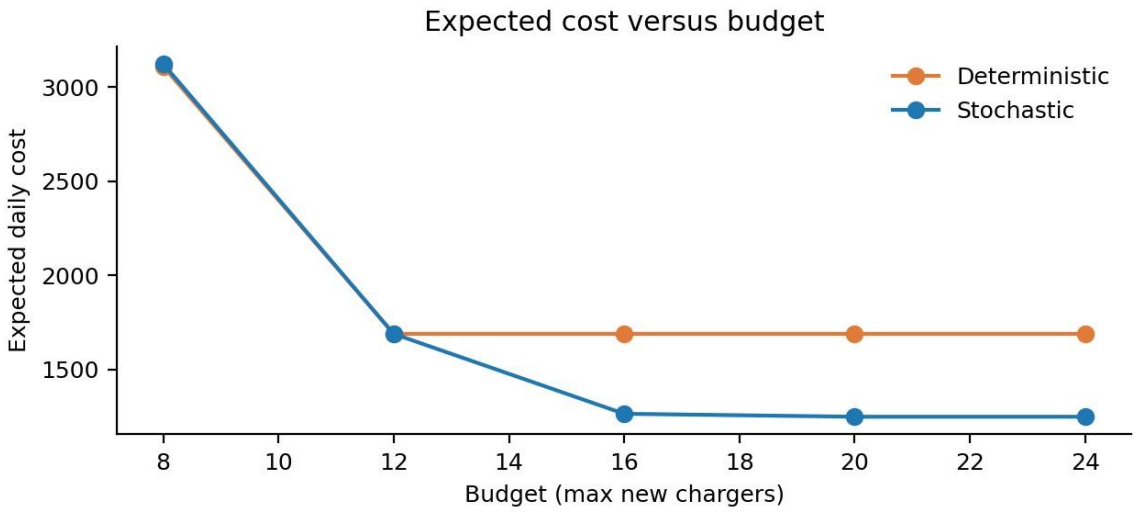


Figure 3. Expected daily cost against the available budget.

With a budget of 8, both plans use all 8 chargers and the VSS is slightly negative (−13.65). This small difference is likely sampling noise, because the stochastic plan was optimized on only 60 scenarios. With a budget of 12, the two plans are identical. For budgets of 16 and above, the deterministic plan stays at 12 chargers because the average-demand model sees no reason to build more, so its cost does not improve. The stochastic plan keeps improving until it uses 16 to 17 chargers, and the VSS rises to about 425–440. The results for budgets 20 and 24 are the same, because the stochastic plan does not spend the whole budget.

6.3 Sensitivity to demand volatility

Next, I varied the zone-level volatility from 0.1 to 0.4 while keeping the budget at 16.

Zone volatility	Deterministic cost	Stochastic cost	VSS	Service level (sto.)
0.1	1,383.09	1,063.60	319.49	99.1%
0.2	1,568.31	1,179.33	388.98	99.2%
0.3	1,822.76	1,364.62	458.14	98.7%
0.4	2,121.74	1,621.39	500.35	97.9%

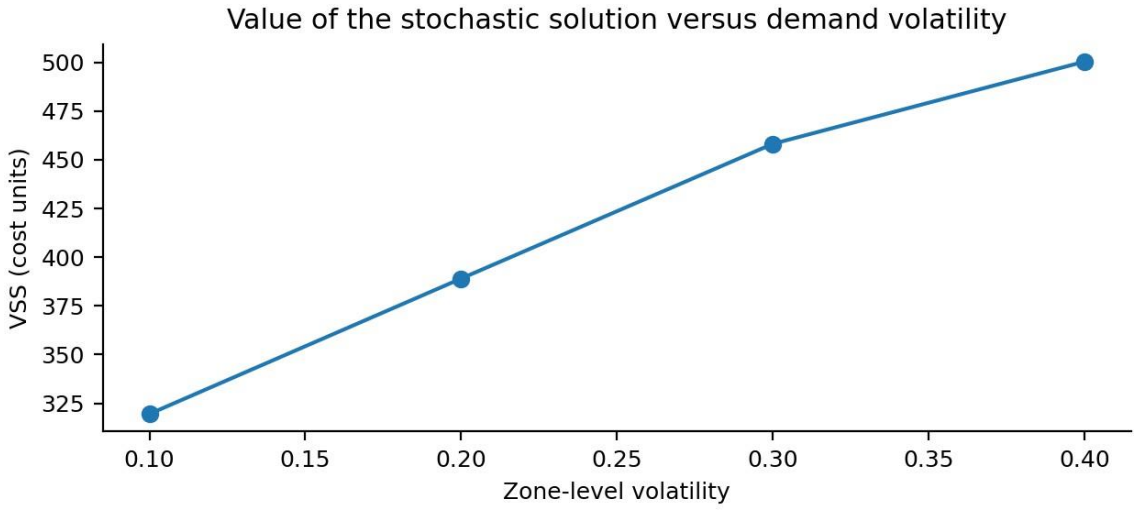


Figure 4. Value of the stochastic solution against zone-level demand volatility.

The stochastic plan always costs less, and the VSS grows with volatility, from 319.49 at 0.1 to 500.35 at 0.4. The higher the uncertainty, the more it is worth planning for it. Even at low volatility the VSS is large, which is probably because the deterministic plan leaves little spare capacity above the mean demand (600 against 570 sessions per day).

6.4 Why compare on separate scenarios?

An optimization model can look better on the scenarios it was trained on than on new ones. Using a separate evaluation set of 1,000 scenarios gives a fairer measurement of how each plan would perform on days it has not seen. It also lets the two plans be compared on exactly the same days.

7. Limitations and conclusion

7.1 Limitations

- Demand follows an assumed lognormal pattern and is not estimated from real charging data.
- The model covers one day only. It ignores time of day, charging duration, queues, and demand growth over the years. • All chargers are identical, and costs, penalties, and travel costs are fixed and linear. Grid and land limits are not included.
- Travel between zones uses straight-line distance and a 6 km limit, not real road networks.
- Only 100 scenarios were used for optimization (60 in the sensitivity runs). The gap between training and evaluation cost and the negative VSS at a budget of 8 show that more scenarios would give more stable plans.
- Existing chargers are fixed, and the model does not decide on removing or relocating them.

For these reasons, the result should be treated as an example of the method rather than a recommendation for a real city.

7.2 Conclusion

This project helped me compare charger-allocation decisions using a two-stage stochastic model instead of relying only on average demand. I built a deterministic plan and a stochastic plan, and tested both on 1,000 fresh demand scenarios. Under the assumptions used here, the stochastic plan lowers the expected daily cost from 1,661.47 to 1,244.70 (VSS = 416.77), increases the service level from 95.2% to 99.0%, and halves the standard deviation of cost. The sensitivity checks showed that the benefit of the stochastic plan grows when demand is more volatile and when the budget allows more chargers. If this model were applied to a real network, the first improvement would be to replace the illustrative assumptions with real charging-session data and real road distances.

References

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3. PuLP documentation: https://coin-or.github.io/pulp/
4. SciPy documentation (scipy.optimize.linprog): https://docs.scipy.org/doc/scipy/
5. NumPy documentation: https://numpy.org/doc/
6. pandas documentation: https://pandas.pydata.org/docs/
7. Streamlit documentation: https://docs.streamlit.io/
8. Course assignment handout, Mini Project — Stochastic Optimization Model, StochasticDecision Making.

Reproducibility note: Results in this report come from the project code (app.py) with the default parameters, random seed 42, 100 optimization scenarios, and 1,000 evaluation scenarios. Different solver versions may choose a different plan when several plans have the same cost. Code repository: [paste your GitHub URL here].